

Examen Coto Precio al 3er. Parcial

1)

$$h = 6380 \text{ Km}$$

$$R_{\text{tierra}} = 6.38 \times 10^6 \text{ m}$$

$$M_{\text{tierra}} = 5.97 \times 10^{24} \text{ kg}$$

$$F_g = F_c \quad m_2 \quad \frac{v^2}{r} = G \frac{M_1 \cdot m_2}{r^2}$$

$$G M m r^2 m v^2 r$$

$$V = G M R + h$$

$$V = (6.67 \times 10^{-11}) (5.97 \times 10^{24}) (6.38 \times 10^6) + 6.380,000$$

$$V = 2.540 \times 10^8 \text{ m/s}$$

$$2) a) \Sigma F = m a_n \quad G \frac{M_{\text{tierra}} m_{\text{sat}}}{r^2} = m_{\text{sat}} \frac{v^2}{r}$$

$$g_0 = \frac{G M_{\text{tierra}}}{R^2 \text{tierra}}$$

$$V = \sqrt{\frac{G M_{\text{tierra}}}{r}} = \sqrt{\frac{g_0 R^2 \text{tierra}}{r}} = \sqrt{\frac{(9.8)(6.37 \times 10^6)^2}{(6.37 \times 10^6) + 300 \times 10^3}} = 7.72 \times 10^3 \text{ m/s}$$

$$a = \frac{v^2}{r} = \frac{(7721.28)^2}{6.37 \times 10^6 \text{ m} + 300 \times 10^3 \text{ m}} = 8.9382 \approx 8.94 \text{ m/s}^2$$

$$a_n = \frac{v^2}{R_{\text{orbital}}} = 0.89 \text{ m/s}^2$$

$$V = \frac{2\pi R_{\text{orbital}}}{T} \Rightarrow T = \frac{2\pi R_{\text{orbital}}}{V} = 5427.70 \text{ s} \approx 1.51 \text{ horas}$$

$$b) W_{\text{realizado}} = \Delta E_c + \Delta E_p = E_{\text{mec final}} - E_{\text{mec inicial}} = E_{\text{orbital}} - E_{\text{superficie}}$$

$$E_{\text{orbital}} = E_{p \text{ orbital}} + E_{c \text{ orbital}} = \frac{G M_{\text{tierra}} m_{\text{satelite}}}{r} + \frac{1}{2} m_{\text{sat}} v_{\text{orb}}^2$$

$$E_{\text{orb}} = - \frac{G M_{\text{tierra}} m_{\text{sat}}}{r} + \frac{1}{2} m_{\text{sat}} \left(\frac{G M_{\text{tierra}}}{r} \right) = - \frac{1}{2} \frac{G M_{\text{tierra}} m_{\text{sat}}}{r}$$

$$E_{\text{orb}} = - \frac{1}{2} \frac{g_0 R^2 \text{tierra} m_{\text{sat}}}{r} = - \frac{1}{2} \frac{9.8 (6.37 \times 10^6 \text{ m})^2 \cdot 1000 \text{ Kg}}{6.37 \times 10^6 + 300 \times 10^3 \text{ m}} = -2.98 \times 10^{10} \text{ J}$$

$$E_{\text{sap}} = E_{p \text{ sup}} = - \frac{G M_{\text{tierra}} m_{\text{sat}}}{R_{\text{tierra}}} = - g_0 R_{\text{tierra}} m_{\text{sat}}$$

$$= -9.8 \text{ m/s}^2 \cdot 6.37 \times 10^6 \text{ m} \cdot 1000 \text{ Kg} = -6.24 \times 10^{10} \text{ J}$$

$$W_{\text{realizado}} = \Delta E = E_{\text{orbital}} - E_{\text{sap}} = -2.98 \times 10^{10} - (-6.24 \times 10^{10}) = 3.26 \times 10^{10} \text{ J}$$

3)

$$m = 1000 \text{ Kg} \quad F = 980 \text{ N} \quad x = 2.8 \text{ cm} \rightarrow x = 0.028 \text{ m}$$

$$T = 2\pi \sqrt{\frac{m_T}{K}} = 2\pi \sqrt{\frac{1100}{35,000}} = \underline{1.11 \text{ s}} \quad \text{período}$$

$$K = \frac{F}{x} = \frac{980}{0.028} = 35,000 \text{ N/m}$$

$$m_T = \frac{W}{g} = \frac{980}{9.8} = 100 \text{ Kg} \quad m_T = 1000 + 100 = 1100 \text{ Kg}$$

$$f = \frac{1}{T} = \frac{1}{1.11} = \underline{0.90 \text{ Hz}} \quad \text{frecuencia}$$

4)

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$d = 0.2 \text{ m} \quad T = \frac{120 \text{ s}}{100}$$

$$I = mgd \left(\frac{T}{2\pi} \right)^2 = (1.8 \text{ Kg}) (9.8 \text{ m/s}^2) (0.2 \text{ m}) \left(\frac{\frac{120 \text{ s}}{100}}{2\pi} \right) = \underline{0.129 \text{ Kg/m}^2}$$